DO GROUPS MAKE WISE DECISIONS?

Daniel Evans – University of Bonn
Georg Schneider – University of Bonn

November 2026

Motivation

- Hiring committees, mutual funds, etc. typically make decisions jointly
- Joint decision-making comes with **benefits** but also **risks:**
  - *Benefits*: access to diverse inputs, deliberation, weighing best inputs
  - *Risks*: redundant or conflicting inputs, herding, overconfident teammates

Motivation

- Hiring committees, mutual funds, etc. typically make decisions jointly
- Joint decision-making comes with **benefits** but also **risks:**
  - *Benefits*: access to diverse inputs, deliberation, weighing best inputs
  - *Risks*: redundant or conflicting inputs, herding, overconfident teammates

$\Rightarrow$ Joint and individual decision-making differ in **several respects at once**

$\Rightarrow$ Difficult to know which features of joint decision-making **enhance decision quality** and which **harm it**

Distinct features of joint decision-making

Joint decision: a shared decision made by a group after discussion

1. **Composition**

2. **Communication**

3. **Collaborative aggregation**

Distinct features of joint decision-making

Joint decision: a shared decision made by a group after discussion

1. **Composition**
   - Number and mix of people in a group (+ inputs they bring)
2. **Communication**
   - Exchange and discussion of inputs
3. **Collaborative aggregation**
   - Agreeing on joint decision instead of aggregating separate judgments

Distinct features of joint decision-making

Joint decision: a shared decision made by a group after discussion

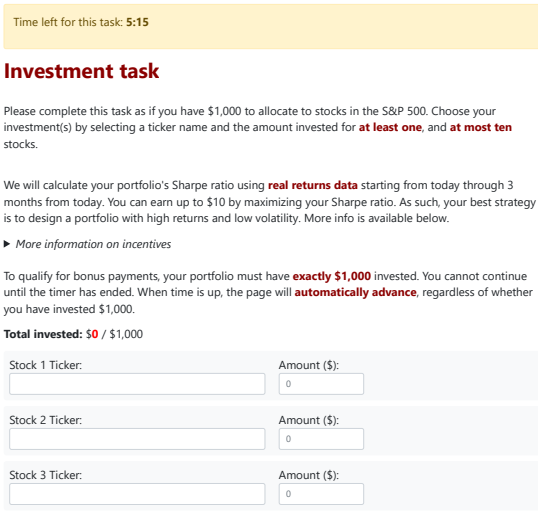
1. **Composition**
   - Number and mix of people in a group (+ inputs they bring)
2. **Communication**
   - Exchange and discussion of inputs
3. **Collaborative aggregation**
   - Agreeing on joint decision instead of aggregating separate judgments

$\Rightarrow$ Need **comparable decisions** & **exogenous variation** in each feature

$\Rightarrow$ Our experiment estimates the **separate contribution** of each one

Design

Main task: portfolio allocation



- Hypothetical $1,000 to invest
- Searchable list of S&P 500 stocks
- Maximize the **Sharpe ratio**
  - Increase returns, reduce risk
- Bonus up to $10 based on Sharpe
- Randomly assigned **decision mode**

Decision modes: visualization

INDIVIDUALS	CROWDS	PANELS	TEAMS
			
Own portfolio	**Pooled portfolio**	Pooled portfolio	**Joint portfolio**
Up to 10 stocks	**Up to 20 stocks**	Up to 20 stocks	Up to 20 stocks
No communication	No communication	**Communication**	Communication
128 individuals	64 pairs	64 pairs	66 pairs

Portfolio aggregation

- *Crowd* and *Panel* aggregation via portfolio **pooling**, e.g.:
  - Subject A: $100 in Stock A, Subject B: $100 in Stock B
  - Pooled portfolio: $50 in Stock A, $50 in Stock B
  - Combining independent inputs can **improve diversification**

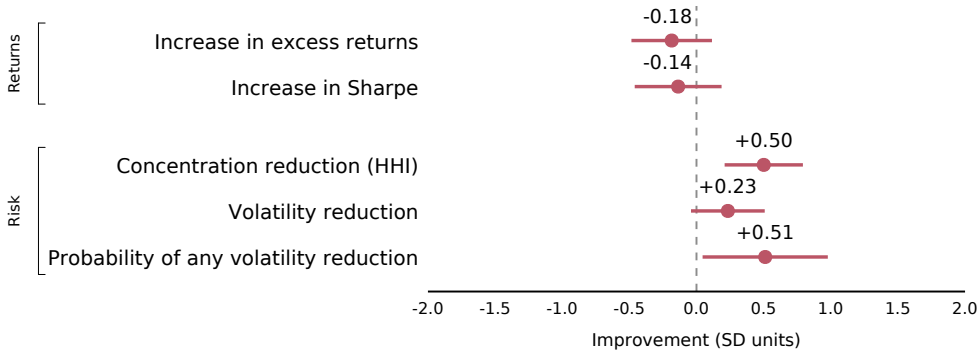
Portfolio aggregation

- *Crowd* and *Panel* aggregation via portfolio **pooling**, e.g.:
  - Subject A: $100 in Stock A, Subject B: $100 in Stock B
  - Pooled portfolio: $50 in Stock A, $50 in Stock B
  - Combining independent inputs can **improve diversification**
- *Teams* aggregate **collaboratively**
  - Shared portfolio input fields, both teammates must agree
  - Not restricted to 50-50 allocation

Results

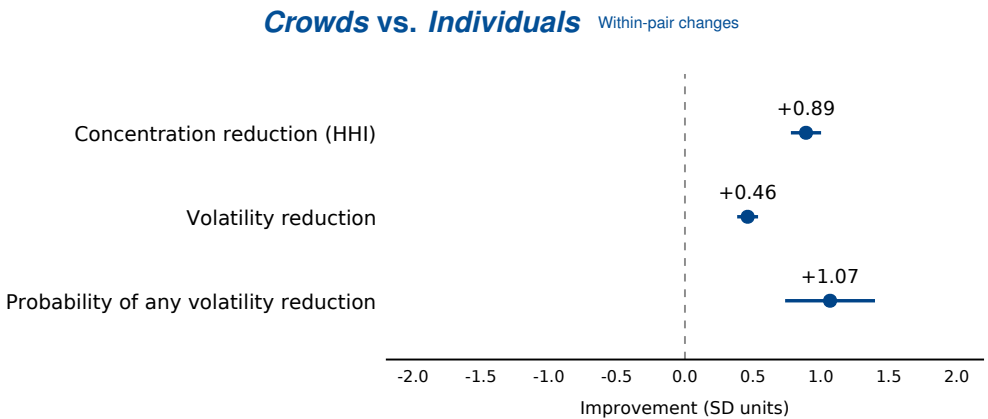
Teams outperform *Individuals* on portfolio risk

Teams* vs. *Individuals, controlling for baseline performance Within-pair changes



- Which features of joint decision-making explain effects on portfolio risk?

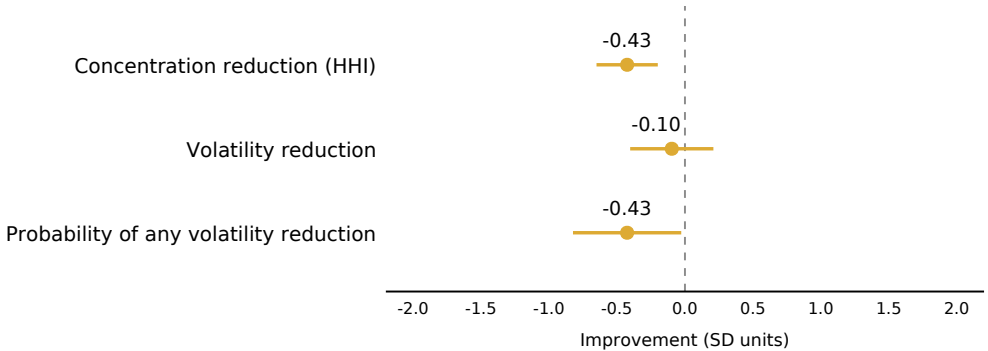
Strong reductions in risk from combining inputs



- Compositional effects **exceed total effect** of joint decision-making

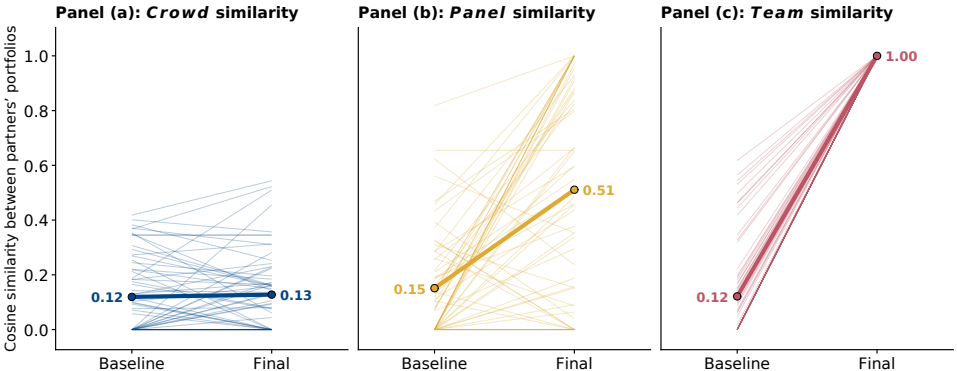
Communication undermines diversification

Panels* vs. *Crowds, controlling for baseline performance Within-pair changes



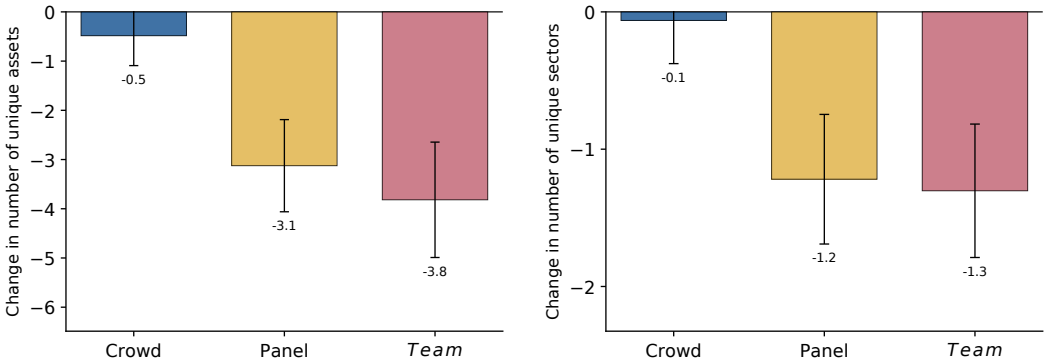
- Communication **attenuates the gains** from combining inputs

Communication makes portfolios more similar



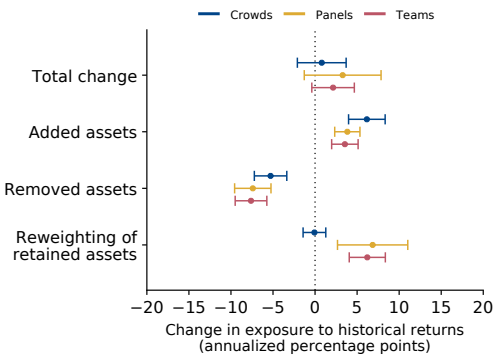
- *Panel* members **converge** despite making **separate decisions**

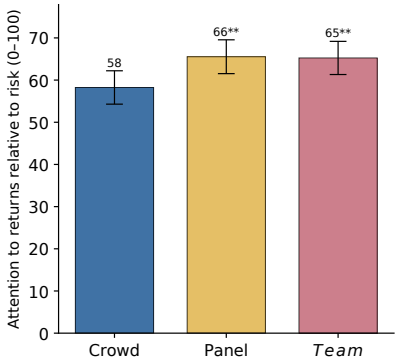
Communication reduces portfolio breadth



- *Panels* & *Teams* **drop unique assets & sectors** held in baseline portfolios

Communication shifts focus to returns

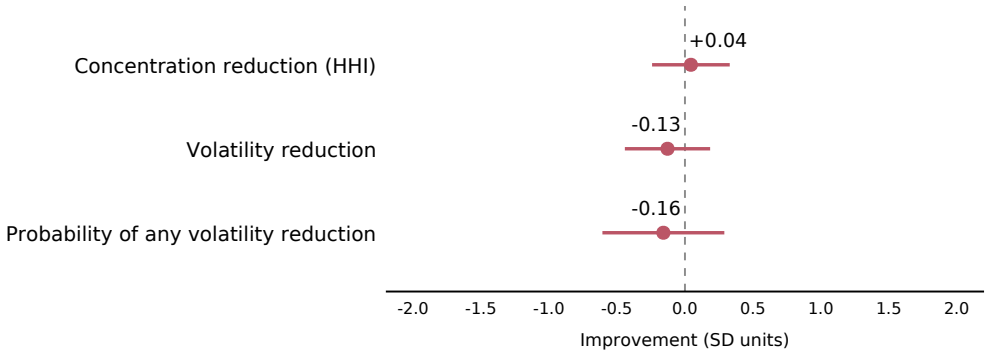




- Members **double down** on stocks with high historical returns held at baseline
- Higher self-reported **focus on returns** (+7–8, both $p < 0.05$)

No average effect of collaborative aggregation

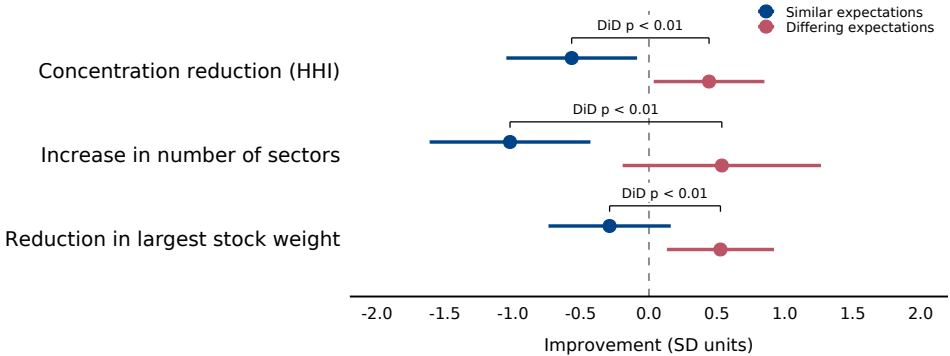
Teams* vs. *Panels, controlling for baseline performance Within-pair changes



- Small and inconsistent effects on portfolio risk

Collaborative aggregation and expectations

Teams* vs. *Panels, controlling for baseline performance Within-pair changes



- Collaborative aggregation improves diversification among belief-diverse pairs

Robustness

Are these results:

1. An inevitable consequence of our design? **No**
   - Subjects know that diversification is a good strategy Evidence
   - Pooling aids diversification & is easy for *Teams* to reproduce
   - *Teams* have larger toolkit: communication and collaborative aggregation

2. Driven by copying of LLM portfolios? **No**
   - Human and LLM portfolios differ systematically Stock choice Breadth

3. Dependent on the evaluation window? **No**
   - Strongest effects in window-invariant measures, e.g. concentration
   - Similar results in alternative evaluation window Evidence

Takeaways

- *Crowds* are the **best-performing** mode on portfolio risk
- *Panel* & *Team* underperformance driven mostly by **communication**, which:
  - Attenuates gains from pooling
  - Shifts focus to returns, but fails to increase them
- Collaborative aggregation can help when groups hold **differing expectations**

Takeaways

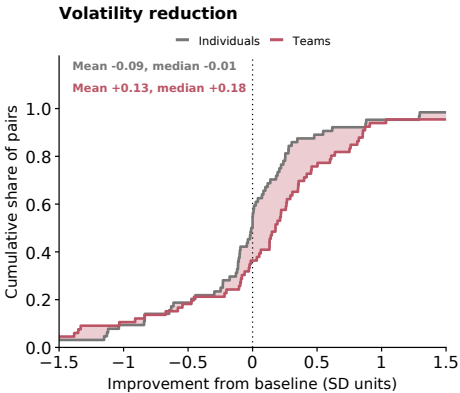
- *Crowds* are the **best-performing** mode on portfolio risk
- *Panel* & *Team* underperformance driven mostly by **communication**, which:
  - Attenuates gains from pooling
  - Shifts focus to returns, but fails to increase them
- Collaborative aggregation can help when groups hold **differing expectations**

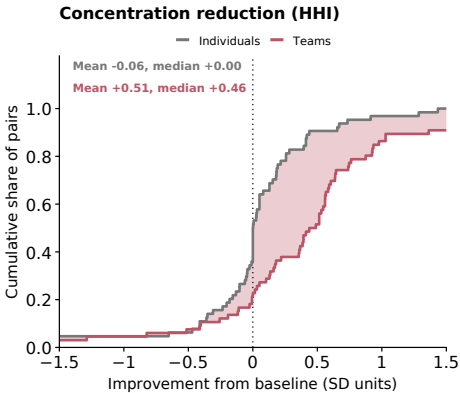
$\Rightarrow$ Decomposition reveals **offsetting effects** that naive comparisons would miss

$\Rightarrow$ Groups with access to diverse inputs may not make make full use of them

Appendix

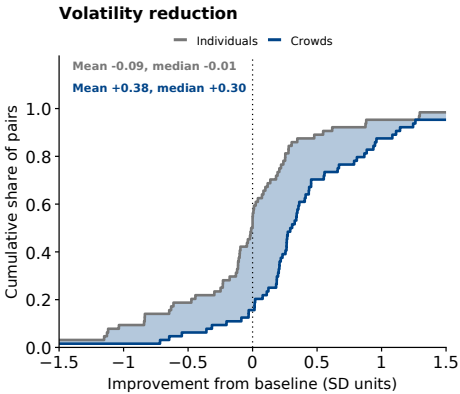
Within-pair changes: *Teams* and *Individuals*

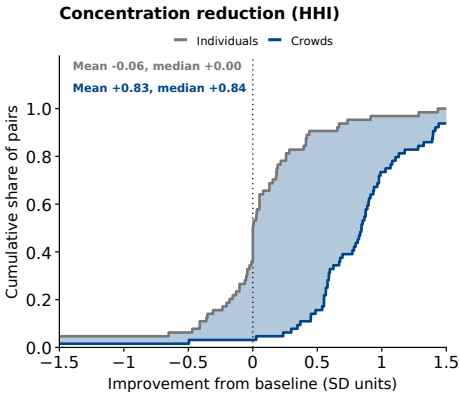




◂ Back to coefficients

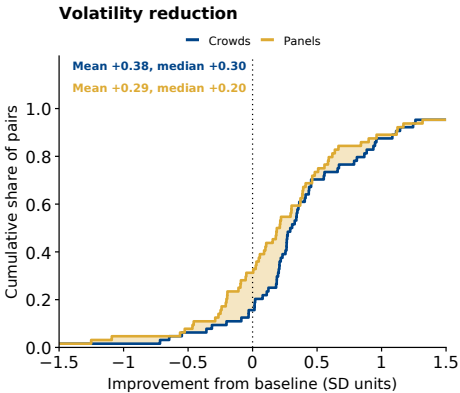
Within-pair changes: Crowds and *Individuals*

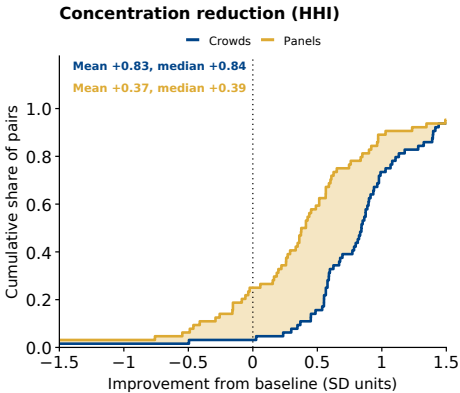




◂ Back to coefficients

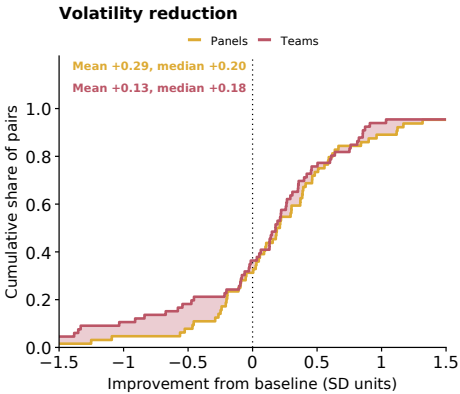
Within-pair changes: Panels and Crowds

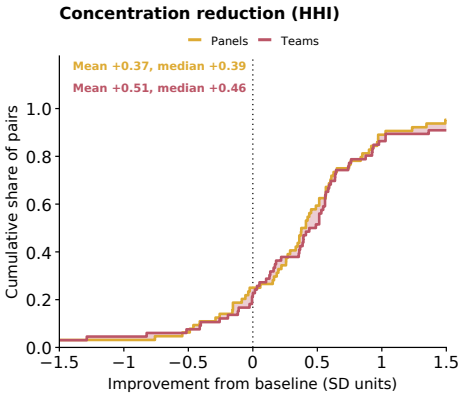




◂ Back to coefficients

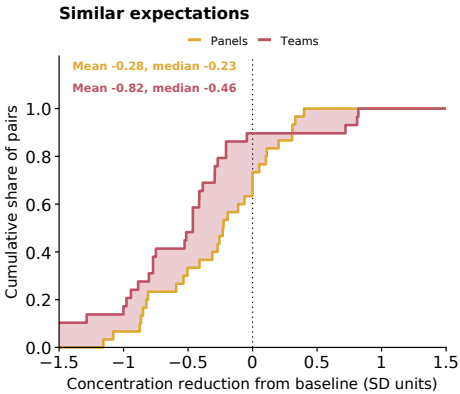
Within-pair changes: *Teams* and Panels

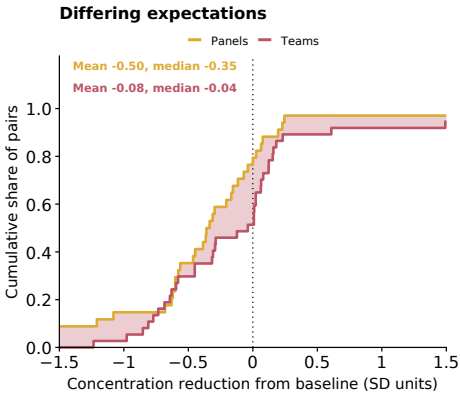




◂ Back to coefficients

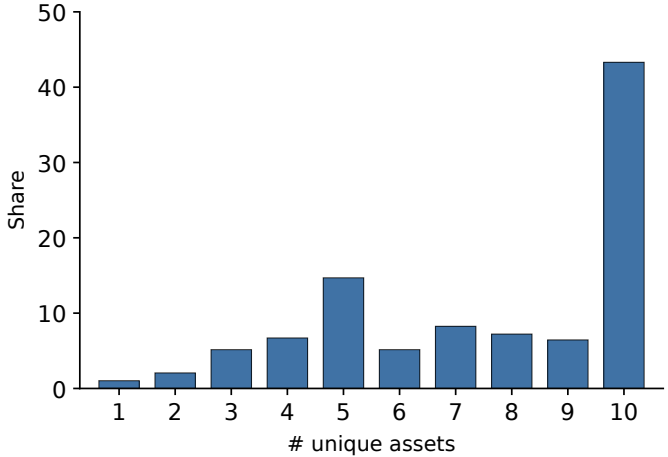
Within-pair changes by expectations





◂ Back to coefficients

Individuals spread across assets

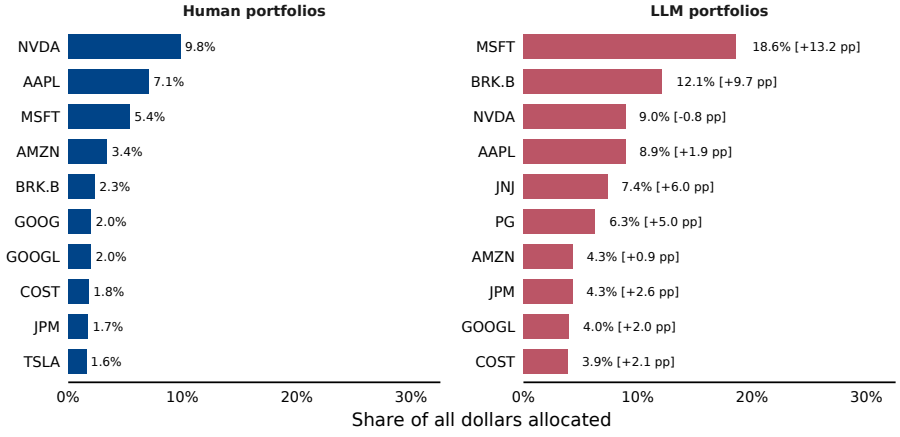


- Mean **7.6** unique assets
- **43%** allocate the maximum of 10

⇒ Attempts to **diversify** in line with incentives

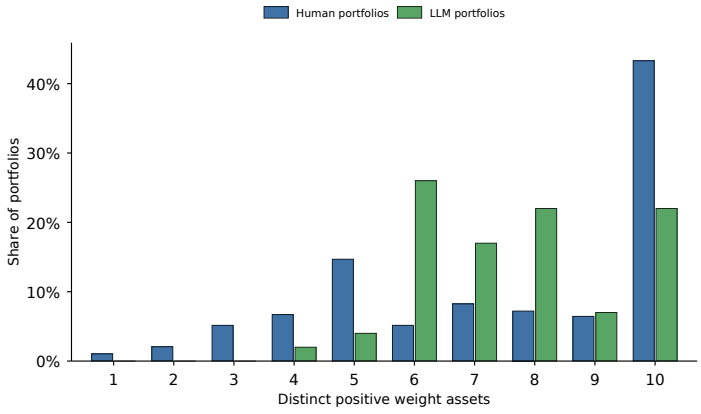
◂ Alternative explanations

Human and LLM stock choices



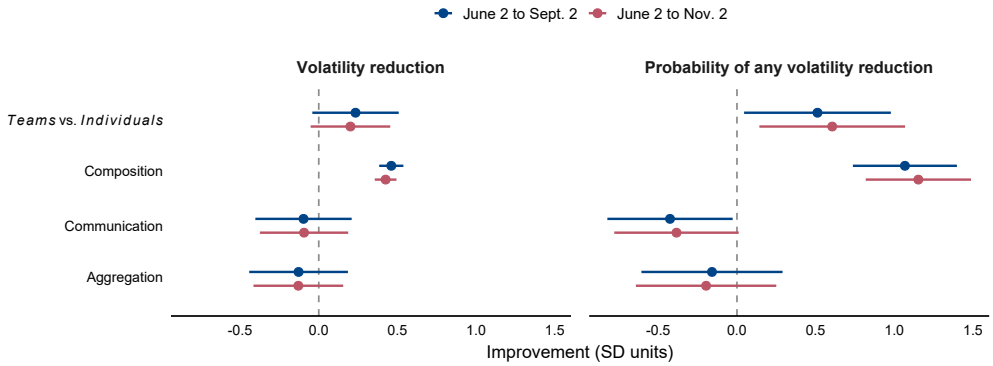
◂ Alternative explanations

Human and LLM portfolio breadth



Alternative explanations

Alternative evaluation window



◂ Alternative explanations